Mohammed ABO ORAIG

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Profile

Engineering graduate in **Data Science and Information Systems** and Master's graduate in **Complex Systems Engineering**, with research experience spanning **machine learning, uncertainty-aware estimation, remote sensing, and mathematical optimization**. Experienced in developing and evaluating data-driven methods for heterogeneous, noisy, and incomplete observations.

Education

Engineering Degree – Data Science & Information Systems Toulon, France
SeaTech Engineering School, University of Toulon (Double-degree program) *Sep 2024 – Sep 2026*
Ranked 1st in the cohort – Grade: 17/20
Relevant coursework: Machine Learning, Statistical Learning, Numerical Optimization, Stochastic Optimization, Scientific Computing.

Master of Science – Complex Systems Engineering (Signal, Vision & Control) Toulon, France
Faculty of Sciences and Technology, University of Toulon (Double-degree program) *Sep 2025 – Sep 2026*
Ranked 1st in the cohort – Grade: 17/20
Relevant coursework: Bayesian Estimation and Inference, Stochastic Modelling, Detection and Estimation Theory, Inverse Problems.

Engineering Degree – Intelligent, Communicating, and Mobile Systems Fez, Morocco
Faculty of Science and Technology of Fez (Double-degree program) *Sep 2022 – Jun 2024*
Ranked 1st in the cohort – Grade: 16/20
Relevant coursework: Digital Signal Processing, Image Processing, Embedded Systems, Numerical Methods.

Research Experience

Research Intern – Multi-Source Trajectory Reconstruction and Robust Estimation
CLS (Collecte Localisation Satellites) Toulouse, France *Mar 2026 – Aug 2026*

- Developed a constrained state-estimation approach to reconstruct ship trajectories from heterogeneous, noisy, and incomplete satellite and vessel-monitoring observations.
- Formulated the reconstruction problem using weighted least squares and implemented a sparse batch estimator in Python using NumPy, SciPy, and LSQR.
- Evaluated estimation accuracy and robustness under missing observations, irregular sampling, and varying measurement quality.

R&D Intern – Machine Learning and Statistical Signal Processing
OSEAN Technology Le Pradet, France *Apr 2025 – Jul 2025*

- Developed and evaluated machine-learning and statistical signal-processing methods to detect and classify marine mammal vocalizations in noisy underwater recordings.
- Applied time-frequency analysis, feature extraction, and dimensionality reduction to derive informative representations from noisy sensor data.
- Compared SVM, LSTM, and stochastic matched-filter approaches, achieving an F1-score of 85% under the evaluation protocol.

Preprints

M. Abo Oraig. *Uncertainty-Aware Multimodal Inference under Sensor Degradation: A Controlled Study of Learned versus Heuristic Sensor Trust. Preprint submitted to arXiv.*

Selected Research Projects

Uncertainty-Aware Multimodal Inference under Sensor Degradation

- Developed an independent research framework for state estimation from heterogeneous sensors under noise, bias, missing observations, and sensor degradation.
- Compared estimation strategies and investigated how unreliable sensor-quality information affects estimation accuracy and robustness.
- Evaluated uncertainty calibration under challenging noise conditions, examining the reliability of confidence estimates under distribution shifts.

Mixed-Integer and Stochastic Optimization for EV Charging

- Formulated EV-charging schedules as constrained optimization problems, accounting for charger availability, time windows, and energy requirements.
- Developed a two-stage stochastic optimization approach to model uncertainty in vehicle arrival times and energy demand.
- Compared stochastic decisions with deterministic forecasts through out-of-sample evaluation, examining the value of explicitly modelling uncertainty.

Learned Regularization for Robust Image Reconstruction

- Investigated image reconstruction under noisy or incomplete measurements and forward-model uncertainty.
- Integrated learned image priors into optimization-based reconstruction and evaluated performance under varying noise levels and measurement degradation.

Generative Deep Learning: Latent Representations and Diffusion Models

- Implemented a variational autoencoder (VAE) in PyTorch to learn compact image representations and investigate the trade-off between latent dimensionality and reconstruction fidelity.
- Implemented a conditional latent-space diffusion model with ancestral and DDIM sampling and classifier-free guidance.
- Conducted controlled ablation studies to examine how latent dimensionality, guidance strength, sampling steps, and sampler stochasticity affect generation quality and sample diversity.

Deep Learning for Object Detection and Image Segmentation

- Implemented computer-vision pipelines using PyTorch and OpenCV for image recognition, object detection, and segmentation.
- Evaluated CNN-based models and approaches including YOLO and the Segment Anything Model (SAM).

Continual Learning for Image Classification

- Investigated continual learning under evolving data distributions, focusing on catastrophic forgetting and the balance between adaptation and retention.
- Implemented and compared EWC, iCaRL, A-GEM, and MIR in PyTorch.
- Evaluated adaptation, forgetting, and generalization across sequential learning scenarios.

Scholarships, Awards & Leadership

Yemeni Ministry of Education National Scholarship 2018–2024
Awarded to top-ranking students nationwide. $36,000

DAFI Scholarship 2022–2024
German government-funded scholarship. €6,000

Project Manager & Co-founder – Dawam Injaz Junior Enterprise Feb 2024 – Jul 2024
Led a multidisciplinary team of 30 engineering students in developing a digital platform, coordinating technical work, project planning, and external stakeholder collaboration.

- Morocco Youth Award 2024 – Third Place €5,000
- Company Program National Competition 2024 – Favorite Award €1,000
- Company Program Regional Competition 2024 – Best Junior Enterprise €1,000

Volunteering

Arabic Wikipedia Editor Oct 2021 – Present
Created and edited 4,500+ articles, improving content accuracy, neutrality, sourcing, and quality across the Arabic-language encyclopedia.

Co-founder & Secretary General – Sicom Club Oct 2023 – Jun 2024
Co-founded and helped lead a university club promoting student learning, innovation, and collaboration.

Languages

English (C1, TOEIC 985/990), **French** (Full professional proficiency), **Arabic** (Native).